Retail Sales & Inventory Intelligence Dashboard

Functional and Non-Functional Requirements

1. Functional Requirements

The functional requirements describe the features and functions that the Retail Sales & Inventory Intelligence Dashboard must provide to help management monitor sales performance, profitability and inventory levels.

ID	Requirement	Acceptance Criteria
FR01	Display total revenue	The dashboard displays total revenue calculated by multiplying the quantity sold by the unit selling price for each sales record and summing the results.
FR02	Display total profit	The dashboard displays total gross profit calculated by subtracting the unit cost from the unit selling price and multiplying the result by the quantity sold.
FR03	Filter sales by category	The user can select a product category using a slicer, and the relevant sales visuals update to reflect the selected category.
FR04	Display sales trends	The dashboard includes a line chart showing revenue over time, allowing management to identify changes in sales performance.
FR05	Identify low-stock products	Products with stock levels at or below their reorder levels are flagged for review.
FR06	Display product sales performance	The dashboard displays sales performance by product, allowing management to compare products based on revenue generated.
FR07	Display units sold	The dashboard displays the total quantity of products sold across the included sales records.
FR08	Filter sales by date	The user can select a date or date range, and the relevant sales visuals update to display results for the selected period.
FR09	Display inventory information	The dashboard includes an inventory table showing each product, its stock on hand, reorder level and stock status.

FR10	Display key performance indicators	The dashboard presents important measures, including total revenue, total profit and units sold, using clear KPI cards.

2. Non-Functional Requirements

Non-functional requirements describe the quality and usability standards that the dashboard should meet.

ID	Requirement	Acceptance Criteria
NFR01	Readability	Charts, tables and KPI cards use clear labels, readable text and appropriate number formats.
NFR02	Consistent formatting	The dashboard uses a consistent colour scheme, font style and layout throughout.
NFR03	Usability	Users can understand the dashboard and interact with its filters without needing extensive instructions.
NFR04	Responsiveness	Dashboard visuals update appropriately when the user changes the category or date filters.
NFR05	Data accuracy	Dashboard calculations and displayed totals agree with independently calculated results from the source data or SQL queries.
NFR06	Maintainability	The dashboard uses clearly named fields and measures so that calculations can be understood and updated when necessary.

3. Project Scope

The project focuses on analysing retail sales and inventory data to support business decision-making. The dashboard is intended to help management monitor revenue, gross profit, product performance and stock levels.

The project uses sample sales and inventory datasets. It does not include live inventory updates, automated purchasing, payment processing or integration with a real retail management system.

4. Test Cases

Test ID	Test description	Test steps	Expected result	Status

TC01	Total revenue card	Compare the card value with the total revenue calculated from the source data.	The card displays the correct total revenue.	Pass
TC02	Total sales card	Compare the total sales card with the source data.	The card displays the correct total sales value.	Pass
TC03	Product sales chart	Check the product sales chart against the source data.	Products and their sales values display correctly.	Pass
TC04	Category slicer	Select a category from the Category slicer.	The relevant charts, tables and cards update to show the selected category.	Pass
TC05	Date slicer	Select a date or date range.	The dashboard displays data for the selected period.	Pass
TC06	Inventory table	Compare StockOnHand and ReorderLevel with the source data.	Inventory quantities display correctly for each product.	Pass
TC07	Stock status	Check products with stock at, below and above their reorder levels.	Each product displays the correct stock status.	Pass
TC08	Slicer interaction	Change the category and date filters together.	The dashboard reflects both selected filters correctly.	Pass
TC09	Dashboard layout	Review the dashboard at normal viewing size.	Titles, cards, charts, tables and slicers are	Pass

			readable and properly arranged.	

Testing Summary

The Sales and Inventory Dashboard was tested to evaluate the accuracy of its calculations, the functionality of its interactive filters, the correctness of its inventory information, and the overall usability of the report. Test cases were designed to verify that the dashboard displays accurate information and responds appropriately to user interactions.

The outcome of each test will be recorded as Pass, Fail, or Not Tested. Any identified issues will be corrected and retested before the dashboard is considered complete.